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May 13, 1975
SUPERSEDING
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June 26, 1964

FEDERAL SPECIFICATION

LANTERN, ELECTRIC, HEAVY-DUTY, (RECHARGEABLE BATTERY)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 Scope. This specification covers a rechargeable battery-operated electric lantern of one type for heavy-duty service.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specification:

PPP-B-636 - Boxes, Shipping, Fiberboard.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC; Atlanta, Chicago, Kansas City, MO; Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks, and the Index of Federal Specifications and Standards, from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

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...e provided for each assembly to prevent unauthorized removal ... detector shall ...

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, NW., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Preproduction sample. When so specified in the invitation for bids, a preproduction sample of the lantern shall be submitted to the contracting officer for test and approval (see 6.2).

3.2 Material. The material used in the construction of the heavy-duty lanterns shall be new and of good commercial quality which is suitable for the purpose intended. All materials used shall be free from defects and imperfections that may adversely affect the serviceability of the finished product. No material shall be used which constitutes a fire hazard.

3.3 Construction. The lantern shall be of sturdy construction, capable of withstanding the strains, jars, vibrations, adverse weather and other similar conditions incident to shipping, storage, and service intended. All parts subject to wear shall be readily accessible for adjustment, repair, or replacement.

3.4 Finish. All external metal parts of the lantern shall be corrosion-resisting or have a corrosion-resisting finish applied by plating or equally effective means. Each lantern unit shall be finished smooth and free of any sharp projections. Surface appearance shall be in accordance with acceptable industrial practice.

3.5 Design, general. The lantern shall be a complete unit, including the lamp, lamphead, rechargeable battery, charger, switch, handle and cord, designed for a minimum of 2.5 hours continuous service at 70°F without recharging. The lantern shall utilize the current drawn from a not less than 4-volt rechargeable lead-acid or 5-volt rechargeable nickel cadmium battery. It shall be capable of being carried by the handle, or of standing on the base without other support. The concentrated light source shall be capable of projecting its beam horizontally without interference from the lantern either when standing or being carried. The switch shall provide a positive instantaneous means of energizing or de-energizing the light source.

3.6 Component parts.

3.6.1 Charger. The charger shall be designed to operate on 115-120 volts, A.C., 60 cycles. The charger shall be furnished as an integral part of the complete lantern assembly with the charging circuit built into the battery case. Charging shall be accomplished by the use of a cheater-type cord furnished with the lantern, which shall mate with a recessed receptacle mounted in the battery case. The charger shall be capable of recharging the battery within 24 hours at 70°F. The charger shall be designed so as to prevent damage to the battery by overcharging. An indicator shall be provided to indicate when the battery has reached a condition of full charge.

3.6.2 Battery. The battery shall be of the rechargeable lead-acid not less than 4 volts, or rechargeable nickel cadmium not less than 5 volts, type. The battery shall be capable of being recharged a minimum of 200 times. The battery shall be of a leakproof, spillproof, nonrefillable maintenance-free construction.

3.6.3 Handle. The lantern shall have a suitable handle for carrying the lantern in the hand. The handle shall distribute the weight of the lantern without excessive strain on any part and shall not interfere with the projected light beam when the lantern is standing on its base.

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3.6.4 Switch, general. A switch shall be provided in the electrical circuit which shall be completely enclosed within the lantern body except for its actuating member. The actuating member shall be on the outside of the lantern unit. If of metal, the exposed actuating member shall be suitably insulated from the electrical circuit to prevent accidental discharge of the battery through contact with any exterior part of the lantern. The switch shall have an "on" and "off" position, constructed to resist accidental movement.

3.6.5 Reflector head assembly, general. The reflector head assembly shall consist of those parts of the lantern which accomplish conversion of the electrical energy of the battery into visible light and the projection of that light as a concentrated beam of light. The essential parts of the assembly shall be a lamp, lampholder, reflector, lens, and assembly casing.

3.6.5.1 Concentrated light, reflector head assembly. Lanterns shall be equipped with a prefocused light source of the sealed beam type, with integral reflector, lens, filament, and screw-type terminals, trade No. 4547, PAR-36, 4.75 volts, 1.25 amperes, or trade No. 4508, PAR-36, 4.0 volts, 1.50 amperes, as applicable. Alternative PAR-36 sealed beam light sources may be used, provided the rated candlepower is not less than 17,000 and the service life is not less than 100 hours.

3.6.5.2 Reflector head assembly movement. The reflector head assembly shall be capable of being rotated a full 90 degrees from a horizontal to a vertical position when the lantern is resting of the base. A stationary head will be permissible if other means are provided to accomplish the same results.

3.6.6 Base. The lanterns shall have a suitable base which will provide a stable platform when subjected to test under the conditions stated in 4.5.2. The base is defined as that surface which is intended by the manufacturer to serve as the support or bottom of the lantern.

3.6.7 Size. The overall dimensions of the complete lantern assembly shall not exceed 13 inches in height, 5 inches in depth, and 8-1/2 inches in length or width.

3.6.8 Weight. The total weight of the complete lantern assembly shall not exceed 7 pounds.

3.6.9 Instruction book. An instruction book shall be furnished with each complete lantern unit. It shall be packed in the shipping carton and shall contain, as a minimum, the general description of the unit, complete instructions for the care and maintenance of the battery, and parts identification by style, model, or catalog number.

3.7 Wiring and connection.

3.7.1 Current-carrying parts. Internal current-carrying parts shall be of brass or phospor bronze or other suitable metal with the proper cross-sectional area for the purpose intended. Bare fixed conductors shall be securely held in position by rivets or other suitable means and properly insulated from contact with any metallic part of the outside of the lantern. Eyelet rivets or other fasteners shall not be used on current-carrying parts if they produce openings to the interior of the battery container or reflector head assembly.

3.7.2 Lamp lead wires. Lamp lead wires shall be provided with permanently attached flanged tongue-type terminals of suitable size for connection to the lamp terminals. Lamp lead wires shall be of ample length of permit replacement of parts and to permit normal movement of lantern parts without binding, chafing, or rubbing.

3.7.3 Circuits. The lantern electrical circuits shall be designed so that it shall be impossible to energize the lamp by making contact between any two points on the exposed exterior surface of the lantern when subjected to the test described in 4.5.4.

3.7.4 Chafing protection. All openings in battery container or reflector head assembly casing through which lead wires pass shall be smooth and well rounded or be provided with grommets or other suitable means to prevent chafing.

3.8 Performance.

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3.8.1 Light characteristics, concentrated light. The illuminating capacity of the light beam shall be equal to the rated candlepower of the PAR-36, trade No. 4547 lamp or of the PAR-36, trade No. 4508 lamp. The light beam produced by the concentrated light reflector head assembly shall consist of essentially parallel rays. When projected onto a screen, 5 feet (1.524 meters) distance from the face of the lens, the concentrated beam shall not be greater than 9 inches (228.6 millimeters) in diameter. Conformance of the light characteristics shall be determined under the conditions stated in 4.5.6.

3.8.2 Switch. The switch shall be capable of completing 25,000 cycles of operation without breakage or apparent wear, or becoming electrically inoperative. A cycle shall be from "off" to "on" position and back to "off" (see 4.5.1).

3.8.3 The assembled lantern shall be capable of standing without other support and shall not tip, fall, or roll when placed on its base on a plane surface inclined 15 degrees from the horizontal in the positions stated in 4.5.2.

3.8.4 Rigidity. While standing on its base on a horizontal plane, the lantern shall be capable of withstanding a compressive load of 50 pounds in accordance with the requirements of 4.5.3.

3.8.5 Workmanship. Workmanship shall be of such quality as to assure procurement of high quality lanterns, free from defects which will affect the appearance of the lanterns or impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling for lot acceptance.

4.2.1 Lot. All lantern units of the same type presented at one time shall be considered a lot for purposes of inspection and tests.

4.2.2 Sampling for inspection. Sampling for inspection shall be in accordance with MIL-STD-105, Inspection Level II, Acceptable Quality Level (AQL) 2.5.

4.2.3 Sampling for tests. A representative random sample shall be taken from the lot in accordance with table I. Acceptance or rejection of the lot shall be based on the number of defective lanterns found in the sample selected and tested as being equal to or less than the specified acceptance number; or equal to, or greater than, the specified rejection number of table I.

TABLE I. Sampling for test

Lot Size	Sample Size	Acceptance Number	Rejection Number
500 and under	3	0	1
501 to 1300	4	0	1
1301 to 3200	5	0	1
3201 to 8000	7	0	1
8001	10	0	1

4.3 Inspection. Each of the sample lanterns selected in accordance with 4.2.2 shall be visually and dimensionally inspected in accordance with MIL-STD-105 by the Government inspector, to verify compliance with this specification. Each sample shall be inspected for workmanship, material, general design, assembly design where applicable, insulation, operation of switching mechanism, dimensions, finish, and markings. Any lantern in the sample having one or more defects shall be rejected, and if the number of defective lanterns in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

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4.4 Lot acceptance test. Each of the sample lanterns selected in accordance with table I shall be subjected to the tests in 4.5 in any order which is satisfactory to the Government inspector. Any sample lantern which does not meet the requirements of all required tests shall cause rejection of the lot represented by the sample. At the discretion of the Government representative, rejected lots may be resubmitted for retesting in accordance with the procedures specified in MIL-STD-105.

4.5 Tests.

4.5.1 Switch test. The lantern shall have its switch operated in a normal manner at a rate not to exceed 20 cycles per minute for a total of 1,000 cycles. A cycle shall be from "off" to "on" positions and back to "off". Mechanical defects or faulty electrical operation of the switch shall constitute failure of the test. Lamps and batteries shall be replaced as required during the test. Failure of the lamps or batteries shall not be grounds for rejection.

4.5.2 Stability test. The lantern shall be placed on a plane inclined as stated in 3.8.3 and shall be free standing when placed in any of seven different equally spaced positions about an axis perpendicular to the plane of the inclined surface. The surface of the plane shall be rough enough to prevent sliding.

4.5.3 Rigidity test. The lantern shall be subjected to an evenly distributed compressive load of 50 pounds applied vertically without shock to the top of the lantern. The load shall remain for 5 minutes. Examination shall disclose no permanent deformation or breakage of parts as a result of the test.

4.5.4 Drop test. The lantern shall be dropped six times from a height of 3 feet into a box of loose, dry sand having a minimum depth of 1 foot and resting on a stable horizontal base. The lantern shall fall three times each on the base and top. No permanent deformation, breakage, except lens, or operating difficulty shall occur as a result of this test. The surface of the sand shall be leveled after each drop.

4.5.5 Circuit test. The lantern, with the switch in the "off" position, shall be placed in all possible positions on a flat steel surface not less than 12 by 12 inches in area. The lamp shall not be energized as a result of this test.

4.5.6 Concentrated light beam test. The lantern shall be tested for compliance with the light performance requirements of 3.8.1. The test shall be conducted in a dark room and the lantern shall be equipped with a fully charged battery. The concentrated light beam shall be projected onto a screen, whose plane is perpendicular to the optical axis of the light assembly, at a distance of 5 feet (1.524 meters) from the lens. The concentrated beam shall not be greater than 9 inches (228.6 millimeters) in diameter. The beam projected on the screen shall be relatively free of all shadows and filament images.

4.6 Inspection of preparation for delivery requirements. An inspection shall be made to determine whether the packaging, packing, and marking comply with the requirements of section 5 of this specification. Defects shall be scored in accordance with table II. For inspection of interior packaging, the sample unit shall be one shipping container fully prepared for delivery, selected at random just prior to the closing operations. Sampling shall be in accordance with MIL-STD-105. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 with an AQL of 4.0 defects per hundred units.

TABLE II. Classification of defects

Examine	Defects
Marking	Omitted; incorrect; illegible; improper size, location sequence, or method of application.
Material	Any component missing, damaged, or not conforming to the applicable specification or requirements stated herein.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling, or distortion of container.

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TABLE II. Classification of defects (con.)

Examine	Defects
Contents	Quantity per container is more or less than required. Note: If the order quantity is less than the standard pack multiple, the container quantity required shall equal the order quantity. Gross weight exceeds the requirements of the box specification.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C, as specified (see 6.2).

5.1.1 Level A. Each complete lantern shall be packaged in a close-fitting box conforming to PPP-B-636, class weather resistant. The weather resistant box shall be closed and waterproof sealed in accordance with the appendix to PPP-B-636.

5.1.2 Level B. Each complete lantern shall be packaged as specified in 5.1.1, except that all of the boxes shall conform to PPP-B-636, class domestic, and the outer boxes shall be closed in accordance with Method II of the appendix to PPP-B-636.

5.1.3 Level C. Each complete lantern shall be packaged to insure protection during shipment and safe delivery to destination.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Unless otherwise specified (see 6.2), four complete lanterns packaged as specified in 5.1 shall be packed in a close-fitting box conforming to PPP-B-636, class weather resistant. Close, waterproof, seal, and strap the boxes in accordance with the appendix of PPP-B-636.

5.2.2 Level B. Four complete lanterns, packaged as specified in 5.1, shall be packed in a close-fitting box conforming to PPP-B-636, class domestic. Closure of boxes shall be in accordance with Method II of the appendix to PPP-B-636.

5.2.3 Level C. The complete lanterns packaged as specified in 5.1 shall be packed to comply with the National Motor Freight Classification rules or the Uniform Freight Classification rules.

5.3 Marking. In addition to any special marking required by the contract or order, all marking shall be in accordance with 5.3.1 or 5.3.2, as specified (see 6.2).

5.3.1 Civil agencies. Marking shall be in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. Marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The lanterns are intended for use in the illumination of objects within a half-mile distance in general heavy-duty outdoor use.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Preproduction sample, if required (see 3.1).
(c) Level of packaging and packing required (see 5.1 and 5.2).
(d) Whether civil or military marking is required (see 5.3).
(e) When other than four units shall be packed per box for level A (see 5.2.1).

6.3 It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples will be necessary prior to award to determine compliance with this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically requested in the invitation for bids and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

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6.4 Federal specifications do not include all types, classes, and sizes of the commodity indicated by the title of the specification, or which are commercially available, but are intended to cover the kind which are suitable for Federal Government requirements.

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See Section 2 of this specification to obtain extra copies and other documents referenced herein. Price 30 cents each.

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